

SECTION 10

Heaps' Law Artifact Investigation

How vocabulary growth was measured in the Bengali corpus, and why the very first chunk spikes with new words

1. What Heaps' Law Predicts

Heaps' Law is an empirical relationship in corpus linguistics stating that the number of distinct (unique) words V in a document grows as a power of the total word count N , following $V = k \cdot N^\beta$, where k and β are corpus-specific constants (typically $0 < \beta < 1$). In plain terms: the more text you read, the more new words you encounter, but at a shrinking rate, since common words repeat while rare words appear only occasionally. This section documents how that growth curve was measured for the Bengali corpus and investigates an anomaly, or "artifact," that appeared in the data.

2. Step-by-Step Process

- 1. Chunk the corpus** — The full list of cleaned Bengali word tokens (17,586,583 words total) was split into sequential chunks of 10,000 words each, processed in the order they appear in the corpus.
- 2. Track seen vocabulary** — A running set, `seen_words`, was maintained across all chunks. For every word in a chunk, the code checked whether it had already appeared; if not, it was added to `seen_words` and counted as a new unique word for that chunk.
- 3. Record cumulative N and V** — After each chunk, the cumulative word count (N) and cumulative unique-word count (V = size of `seen_words`) were appended to two tracking lists, building the raw data points for the Heaps' curve.
- 4. Flag artifacts** — A chunk was flagged as an artifact whenever its count of brand-new unique words exceeded 35% of the chunk size — an unusually high novelty rate for natural text, where repetition normally dominates once a corpus is underway.
- 5. Log the artifact** — For any flagged chunk, the cumulative N , the number of new words, and a 15-word sample of those new words were printed for manual inspection.
- 6. Plot the growth curve** — Once all chunks were processed, cumulative N was plotted against cumulative V , producing the Heaps' Law curve shown in Figure 1.
- 7. Fit the power law** — $\log(N)$ and $\log(V)$ were computed and fit with linear regression (`numpy.polyfit`) to a straight line. The slope of that line is β , and the intercept exponentiated gives k — since $V = k \cdot N^\beta$ becomes $\log(V) = \log(k) + \beta \cdot \log(N)$ in log-space.

3. Results

Metric	Value
Total word count (N)	17,586,583
Unique word count (V)	607,534
Chunk size used	10,000 words / chunk
Artifact threshold	> 35% new unique words in a chunk
Artifact detected at	$N = 10,000$ (first chunk) — 3,974 new words (39.74%)
Fitted Heaps' Law: $V = k \cdot N^\beta$	$k = 10.9928, \beta = 0.6534$

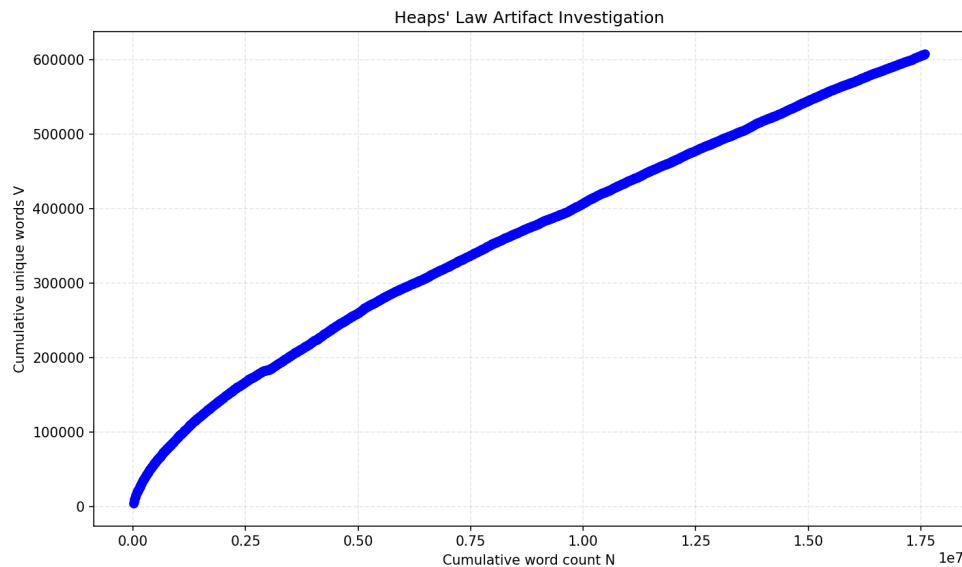


Figure 1: Cumulative unique words (V) vs. cumulative total words (N) for the Bengali corpus.

4. Interpreting the Artifact

- **Only one artifact was found** — at $N = 10,000$, the very first chunk processed. Of its 10,000 words, 3,974 (39.74%) were new unique words, comfortably above the 35% threshold.
- **Why it happens at the start:** `seen_words` begins empty, so every word in the first chunk is, by definition, "new." This is a cold-start effect rather than a genuine irregularity in the language — the very first slice of any corpus will always show an inflated novelty rate simply because there is no prior vocabulary to compare against.
- **No further artifacts:** as the chunks progress, previously-seen high-frequency words (function words, common nouns, grammatical markers) begin dominating each new chunk, so the new-word rate naturally drops well below 35% for the rest of the corpus — visible in Figure 1 as the curve's rapid early rise followed by a steady, decelerating climb.
- **Reading $\beta = 0.6534$:** this falls at the higher end of the typical 0.4–0.6 range reported for English corpora, consistent with Bengali's morphologically rich, agglutinative structure — inflection and compounding generate many surface word-forms from the same root, so unique-word count grows faster relative to total word count than it would for a more analytic language.

5. Conclusion

The chunk-wise scan of the Bengali corpus confirms Heaps' Law holds well overall ($V = 10.99 \cdot N^{0.6534}$), with vocabulary growth decelerating smoothly as expected. The single detected artifact at the very first chunk is not a data-quality problem — it is the expected cold-start signature of the `seen_words` tracking method, where the initial chunk cannot help but register 100% of its words as "new." Beyond that first chunk, no further anomalies were found, indicating the corpus is consistent and free of irregular vocabulary spikes (e.g., boilerplate text or encoding artifacts) later in the stream.